Ferdinand Gotthold Max Eisenstein — Ableitung des biquadratischen Fundamentaltheorems aus der Theorie der Lemniscatenfunctionen (1846/47)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 17 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Ferdinand Gotthold Max Eisenstein of Berlin (1823–1852), the prodigy Gauss championed and Humboldt supported, who signed this memoir at twenty-two and died of tuberculosis at twenty-nine, Humboldt following his coffin.

The author's own terms

φ(t), his lemniscatic sine, satisfying $\partial\varphi(t)/\partial t = \sqrt{(1-\varphi(t)^4)}$. *m* primary, in Gauss's sense: $m = a+bi \equiv 1 \pmod{2+2i}$; $p = N(m) = a^2+b^2$. [n/m], his bracket symbol: the biquadratic residue character, the unit to which n^((p−1)/4) is congruent mod m. *Viertelrestensystem*, the quarter residue system: Gauss's partition of a residue system into quarters by the units. A *ganze Function* is a polynomial, a whole function in his period sense, not the modern entire function. His ∂ is kept as written.

The story

Ferdinand Gotthold Max Eisenstein was born in Berlin on 16 April 1823. He

was not born in Danzig, as the brief for this build had it; the folder corrects that error, and Berlin stands corroborated. At the Friedrich Wilhelm Gymnasium from 1837, then at the Friedrich Werder Gymnasium, he outran the syllabus by fifteen; at seventeen he attended Dirichlet's lectures while still at school.

In 1843 Hamilton gave him Abel's paper on the impossibility of solving the quintic, turning him toward research. Crelle refereed his first papers and brought him to Humboldt, his patron. He visited Gauss at Göttingen in June 1844, and habilitated at Berlin in 1847. Gauss's verdict, as the folder quotes it, was talents “such as nature bestows on only a few in each century” (14 April 1846, to Humboldt).

In the Berlin fighting of 19 March 1848 he was removed from a house from which shots had been fired, taken to the Spandau citadel, and mistreated; released the next day, he never fully recovered. MacTutor records his election to the Göttingen Academy in 1851 and to the Berlin Academy early in 1852. After a haemorrhage in late July 1852 he died in Berlin on 11 October 1852, aged twenty-nine.

The work and what survives

The memoir, *Beiträge zur Theorie der elliptischen Functionen* I, signed “Im October 1845,” appeared in the *Journal für die reine und angewandte Mathematik* 30 (1846), pp. 185–210, and was collected in *Mathematische Abhandlungen* (Reimer, 1847), pp. 129–154. §§1–3 move from multiplication formulas through the quarter residue system to the biquadratic residues and the reciprocity law; §§4–5 compute the multiplication coefficients and give the transformation equations.

From the lemniscatic sine's equation alone, he derives the multiplication formula for an odd Gaussian-integer multiplier and proves every coefficient but one divisible by the multiplier — a theorem which, in his own words, “seems destined to form the foundation for all applications of the division of the lemniscate to number theory” — and extracts the reciprocity law in Gauss's form, correcting a misprint in Jacobi's *Fundamenta nova*.

The folder holds the whole memoir, §§1–5, complete in English draft, made

entire, the folder states, from the digitised 1847 Reimer edition, worked from its page images, ambiguous readings flagged in place. It remains a draft; no claim in it is settled until the reviews below are done. His name survives in Eisenstein integers, Eisenstein's criterion, and Eisenstein series, while the memoirs stand in German.

Its English fortunes

No English translation of this memoir was found after research; its results are restated wherever the early reciprocity laws are taught.

That is the finding at its stated bound. The searches returned German originals and reprints, biographies, and modern reworkings of his results — exposition, not translation. WorldCat and union catalogues, the reciprocity-law histories in full text, and dissertation literature were not searched; passage-level excerpts, if any exist, would narrow the claim. Gauss's *Theoria residuorum biquadraticorum*, in English on Wikisource, is Gauss's Latin memoir, a different author's work; it does not touch this finding.

What completing it would take

Nothing remains untranslated; completing it means verifying the draft. The folder's gates are unmet: review by a number theorist and a period-German reader, with collation against the 1847 page images. The folder defers to a notes file that is not in the folder: the checksums, display image evidence, other-memoirs map, and English-status search record it was to hold cannot be re-verified from the folder, and recovering or rebuilding it is open work. The *Abhandlungen*'s other memoirs await a further word.

Sources

- Project folder: `~/workspace/rare-and-obscure-books/books/eisenstein-reciprocity/eisenstein-reciprocity.txt`
- PIN: `~/workspace/packages/boneyard-extension/research/PIN-eisenstein-reciprocity.md`
- MacTutor, Eisenstein biography: https://mathshistory.st-andrews.ac.uk/Biographies/Eisenstein/

- Wikisource, Gauss's *Theoria residuorum biquadraticorum* — a different author's work, named only to separate it: https://en.wikisource.org/wiki/Translation:Theoria_residuorum_biquadraticum